

GradeSense — Test & Reliability Report

Question used: Q2 — Technology in Education Essay

This report walks through 7 real, live test cases run against the running application — not staged mockups. Five of them are genuine photographs of real handwriting, written by hand and uploaded exactly as a student would, not typed text. Each page shows the actual screenshot(s) GradeSense produced: how it read the transcribed answer, and — for every handwritten case — the real photograph it read it from.

Summary

- | | |
|---|----------------------------|
| 1. Fully Correct Answer (real handwriting) | PASS |
| 2. Partially Correct Answer (real handwriting) | PASS |
| 3. Incorrect Answer (real handwriting) | PASS |
| 4. Blank Answer (real handwriting) | PASS |
| 5. OCR-like Spelling Errors (real handwriting) | PASS (with an honest flag) |
| 6. Unsupported File Upload (extra, not one of the 8 required) | PASS |
| 7. Model / API Failure | PASS |

The remaining 2 of the 8 required categories — malformed/incomplete model output, and a score that would exceed the maximum unless corrected — are not shown here as live screenshots, because neither can be triggered through normal use of the running app: a real API call can't be made to return broken JSON on demand, and the app's own marks-clamping code makes an over-max score unobservable even if the model tried to report one. Both are instead verified by deliberately mocking the grader's response at the code level — see tests #6 and #8 in the automated suite (backend/tests/gradingPipeline.test.ts, output captured in samples/TEST_OUTPUT.md in the repository).

1. Fully Correct Answer

PASS

Why this test matters: Assignment requires a test for "a fully correct answer" — the grader must recognize a genuinely strong answer and award full marks, not just find reasons to dock it.

What we did: Photographed a real handwritten answer for Q2 — a complete essay that takes a clear position, gives a concrete example, honestly addresses the opposing viewpoint, and reaches a coherent conclusion — and uploaded the photo directly, the same way a teacher would receive one.

What happened: AI vision correctly transcribed the handwriting and scored it 5/5 across every rubric point. Still flagged for human review — that's the honest default for any AI-read image, not a sign anything was wrong.

Question 2 — Technology in Education Essay

Student: Raghav Rath (Roll 01) - Subject: English

5 / 5

Reviewed by Teacher
100% model confidence

Student's Answer Model Answer Original Upload

Open in new tab

GradeSense Answer Paper

Student: Raghav Rath - Roll: 01

Subject: English

Date: 9/2/2026

Question 2 — Technology in Education Essay

Name: Raghav Rath

In my opinion technology has made learning Signi Ficantly better for students. Information that used to take hour to find in a library is now avialable within seconds through search engine on line Coureses and educational vidoes. This means Students can explore topics at their own pace and revisit material as many times as they need until they truly understand it. for example as student who struggles with a difficult chemistory concept can watch several different explanation on Youtube until one of them finally makes sense.

However, som epeople argue that this access makes students lazier thinkers since they can simply look up an answer instead of working through a problem themselves. This is a fair concern and is true that overreliance on searching for answer can weaken a student skills.

Even so I believe benefits outweigh this risk as long as technology is used thought fully. A student who uses the internet to understand a concept more deeply rather then to copy a finished answer, endes up learning more. Technology should be treated as a tool not ad replacement.

RUBRIC CRITERIA

SCORE: 5 / 5

- > • States a clear, unambiguous position on the essay prompt 1/1
- > • Provides well-structured arguments supported by relevant concrete examples 1/1
- > • Meaningfully considers and addresses an opposing viewpoint 1/1
- > • Uses relevant examples and demonstrates reasoning rather than merely making unsupported claims 1/1
- > • Concludes with a coherent, logical synthesis of the presented arguments 1/1

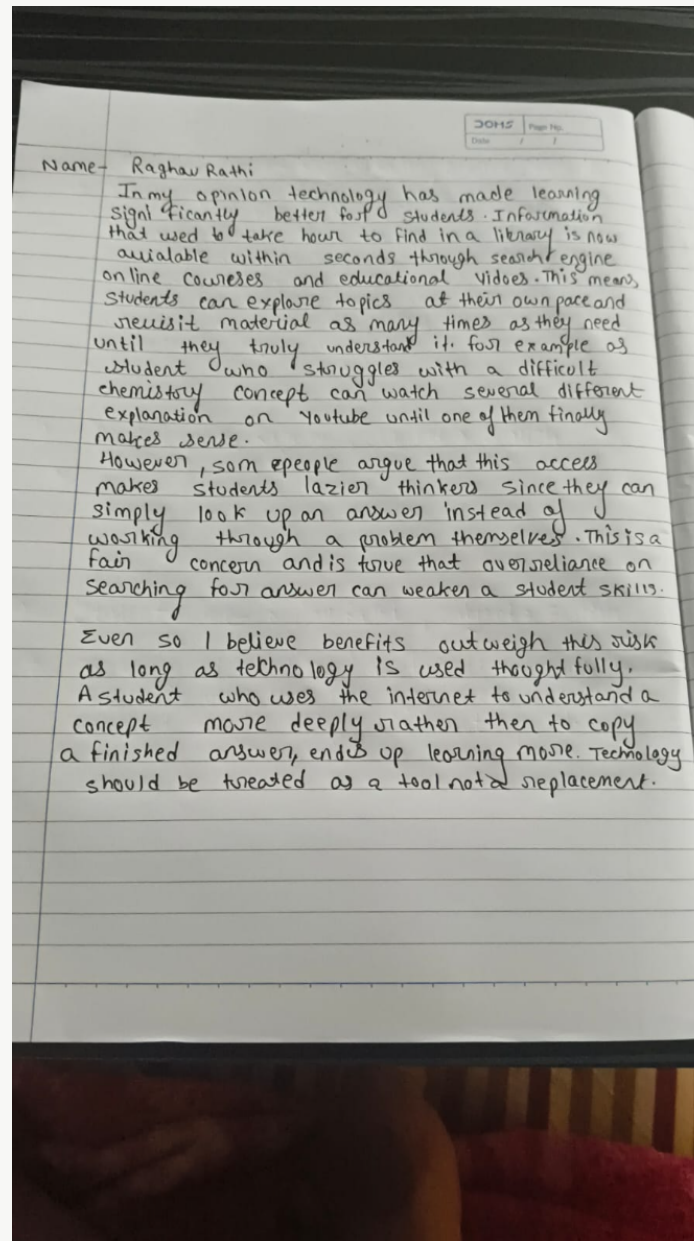
TEACHER NOTES

Free-standing notes you add yourself — not tied to any rubric point or its marks. These appear on their own "Teacher Notes" page in the exported PDF, separate from the rubric breakdown.

+ Add Manual Note

1. Fully Correct Answer — the original handwritten photo

Shown for direct reference — exactly what GradeSense actually read to produce the result on the previous page.



2. Partially Correct Answer

PASS

Why this test matters: Assignment requires a test for "a partially correct answer" — marks should be lost specifically where the answer is actually weak, not as one blunt overall guess.

What we did: Photographed a real handwritten answer that states a position, gives one example, and briefly touches the opposing view — but the argument runs out before a real conclusion, cut off mid-thought, a very realistic case.

What happened: Scored 3/5. Marks awarded for the position, the example, and the opposing-view mention; marks lost specifically on reasoning depth and the missing conclusion — not a flat average.

Question 2 — Technology in Education Essay

Student: **Raghav Rath**i (Roll 01) · Subject: **English**

3 / 5

● Reviewed by Teacher
100% model confidence

Student's Answer

Model Answer

Original Upload

🔗 [Open in new tab](#)

GradeSense Answer Paper

Student: **Raghav Rath**i · Roll: 01

Subject: **English**

Date: 9/2/2026

Question 2 — Technology in Education Essay

JOHS

Date / Page No. 1

Student name - Raghav Rath, subject - English

In my opinion technology has made learning significantly better for students.

Information that used to take hours to find in the library is now available within seconds through search engines, online courses and educational videos. This means students can explore topics at their own pace and revisit material as many times.

However some people argue that this easy access makes students lazier thinkers, since they can look up an answer instead of working hard.

Even so I believe outweigh this risk as long as technology is used thoughtfully and should be used as a tool that supports thinking.

RUBRIC CRITERIA

SCORE: 3 / 5

- States a clear, unambiguous position on the essay prompt 1/1
- Provides well-structured arguments supported by relevant concrete examples 0.5/1
- Meaningfully considers and addresses an opposing viewpoint 0.5/1
- Uses relevant examples and demonstrates reasoning rather than merely making unsupported claims 0.5/1
- Concludes with a coherent, logical synthesis of the presented arguments 0.5/1

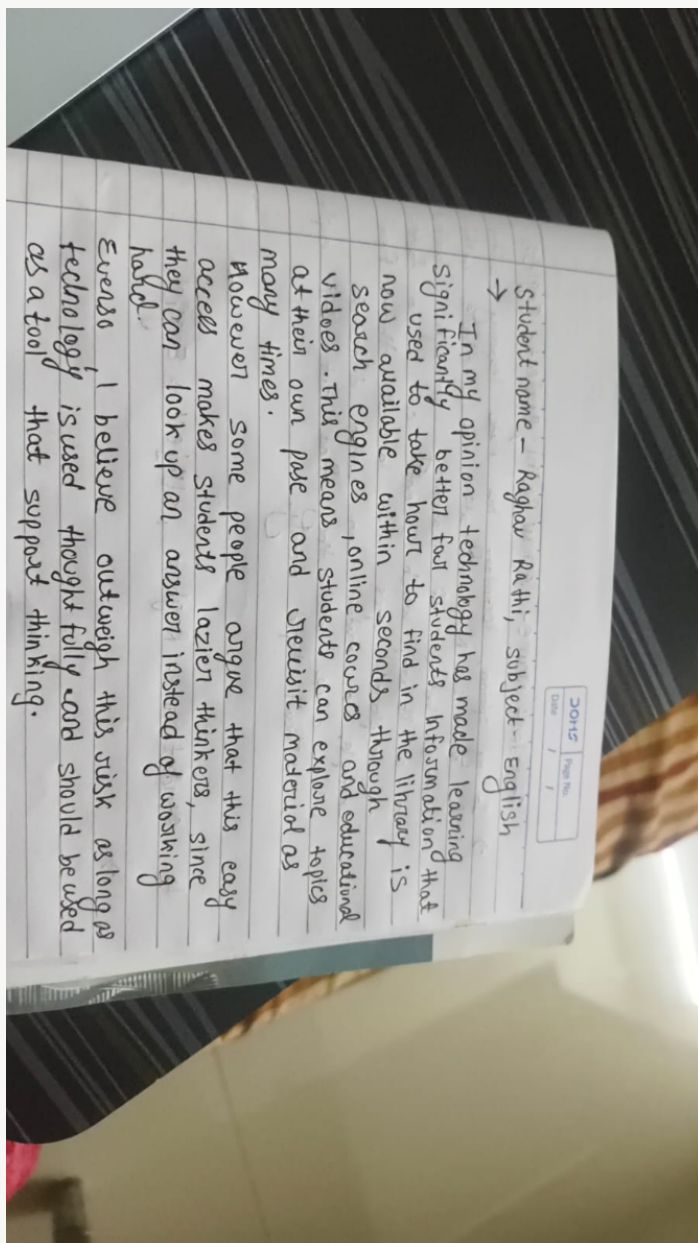
TEACHER NOTES

Free-standing notes you add yourself — not tied to any rubric point or its marks. These appear on their own "Teacher Notes" page in the exported PDF, separate from the rubric breakdown.

+ Add Manual Note

2. Partially Correct Answer — the original handwritten photo

Shown for direct reference — exactly what GradeSense actually read to produce the result on the previous page.



3. Incorrect Answer

PASS

Why this test matters: Assignment requires a test for "an incorrect answer" — a vague, position-less response should not receive undue credit just for being on-topic.

What we did: Photographed a real handwritten answer with no real position and no argument ("Technology is very bad for students... Technology is not good.").

What happened: Scored 0/5. No rubric point found enough substance to award marks for.

Question 2 — Technology in Education Essay

Student: Raghav Rath i (Roll 01) · Subject: English

0 / 5

Reviewed by Teacher

100% model confidence

Student's Answer

Model Answer

Original Upload

Open in new tab

GradeSense Answer Paper

Student: Raghav Rath i · Roll: 01

Subject: English

Date: 9/2/2026

Question 2 — Technology in Education Essay

→ Student name- Raghav Rath i, Subject English

→ Technology is very bad four students and students are also very bad. Technology needs to learn from student as Technology is noot good.

RUBRIC CRITERIA

SCORE: 0 / 5

> States a clear, unambiguous position on the essay prompt

0/1

> Provides well-structured arguments supported by relevant concrete examples

0/1

> Meaningfully considers and addresses an opposing viewpoint

0/1

Uses relevant examples and demonstrates reasoning rather than merely making unsupported claims

0/1

> Concludes with a coherent, logical synthesis of the presented arguments

0/1

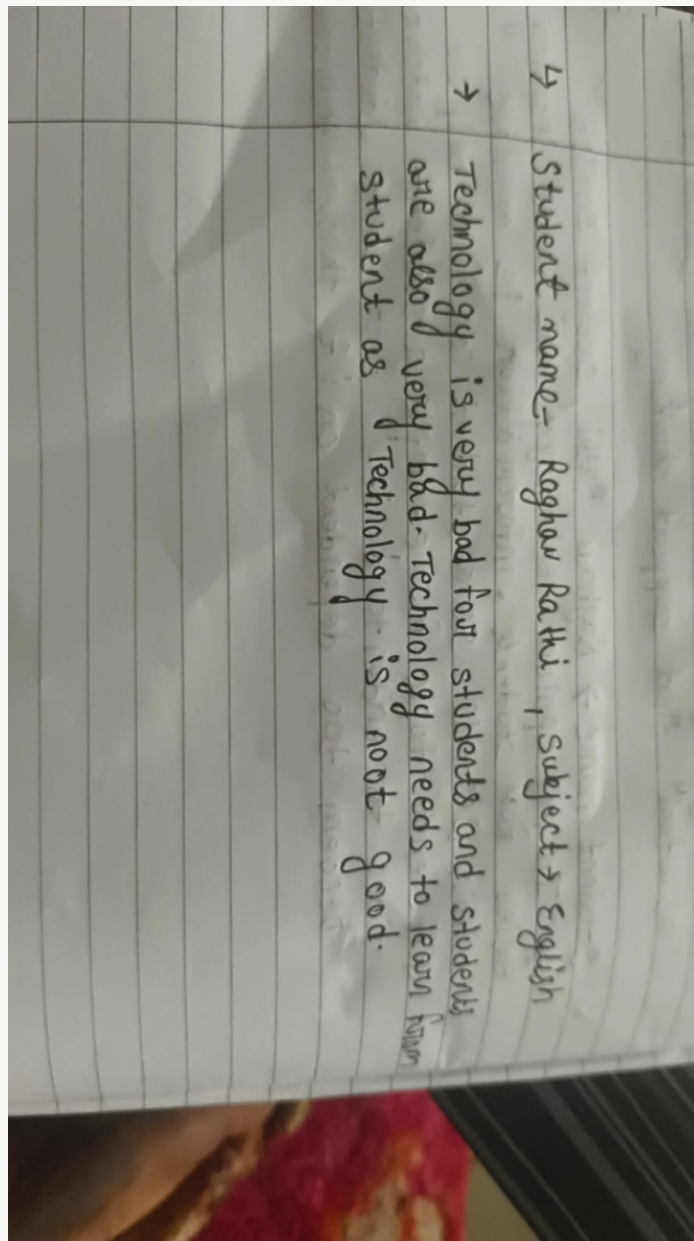
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+ Add Manual Note

3. Incorrect Answer — the original handwritten photo

Shown for direct reference — exactly what GradeSense actually read to produce the result on the previous page.



4. Blank Answer

PASS

Why this test matters: A blank PDF is unambiguous, so the code can safely short-circuit to a confident 0. A photographed blank page is a different case: the AI still has to actually look at the image and confirm there's nothing usable there, so it should be graded honestly rather than either a false short-circuit or a false confident zero.

What we did: Photographed a real answer sheet with only the header filled in — student name and subject — and no actual answer written.

What happened: Scored 0/5 — every rubric point correctly found no relevant content. Flagged for review at reduced confidence, the same as any other AI-read image, exactly as it should be.

Question 2 — Technology in Education Essay

Student: Raghav Rathl (Roll 01) · Subject: English

0 / 5

Reviewed by Teacher

100% model confidence

Student's Answer

Model Answer

Original Upload

[Open in new tab](#)

GradeSense Answer Paper

Subject: English

Student: Raghav Rathl · Roll: 01

Date: 9/2/2026

Question 2 — Technology in Education Essay

Student name - Raghav

subject -> English

RUBRIC CRITERIA

SCORE: 0 / 5

> States a clear, unambiguous position on the essay prompt

0/1

> Provides well-structured arguments supported by relevant concrete examples

0/1

> Meaningfully considers and addresses an opposing viewpoint

0/1

Uses relevant examples and demonstrates reasoning rather than merely making unsupported claims

0/1

> Concludes with a coherent, logical synthesis of the presented arguments

0/1

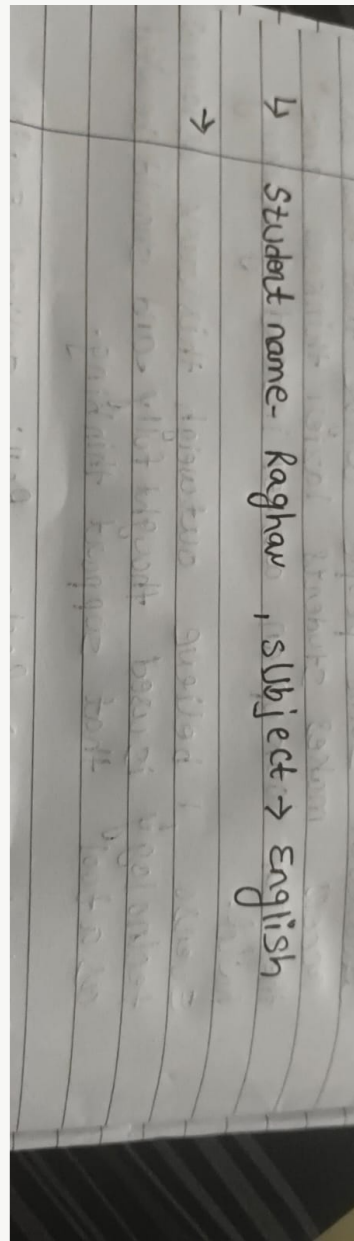
TEACHER NOTES

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+ Add Manual Note

4. Blank Answer — the original handwritten photo

Shown for direct reference — exactly what GradeSense actually read to produce the result on the previous page.



5. OCR-like Spelling Errors

PASS (with an honest flag)

Why this test matters: Assignment requires a test for "an answer with OCR-like spelling errors". With real handwriting this is no longer simulated OCR noise ("vv" for "w") — it's genuine spelling mistakes a real student actually makes, and the grader must still extract meaning from them.

What we did: Photographed a real handwritten answer, deliberately misspelling many words throughout ("oppinion", "diffrent", "becoem", "untill") while keeping the underlying argument coherent.

What happened: AI vision correctly transcribed every misspelling exactly as written, and scored it 2.5/5 based on the actual argument, which only partially covers the rubric — proving spelling noise neither cost nor saved any marks; only the substance did.

Question 2 — Technology in Education Essay

Student: Raghav Rathī (Roll 01) · Subject: English

2.5 / 5

Reviewed by Teacher

100% model confidence

Student's Answer

Model Answer

Original Upload

[Open in new tab](#)

GradeSense Answer Paper

Subject: English
Date: 9/2/2026

Question 2 — Technology in Education Essay

→ Student name → Raghav Rathī, subject English
→ In my opnion technology has made learning easier for students. Information is available within seconds so a student can find help on diffrent topic. for example a student watcha video to undertand it. But some student becoem too dependent on it through them selve.

RUBRIC CRITERIA

SCORE: 2.5 / 5

> States a clear, unambiguous position on the essay prompt

1/1

> Provides well-structured arguments supported by relevant concrete examples

0.5/1

> Meaningfully considers and addresses an opposing viewpoint

0.5/1

Uses relevant examples and

> demonstrates reasoning rather than merely making unsupported claims

0.5/1

> Concludes with a coherent, logical synthesis of the presented arguments

0/1

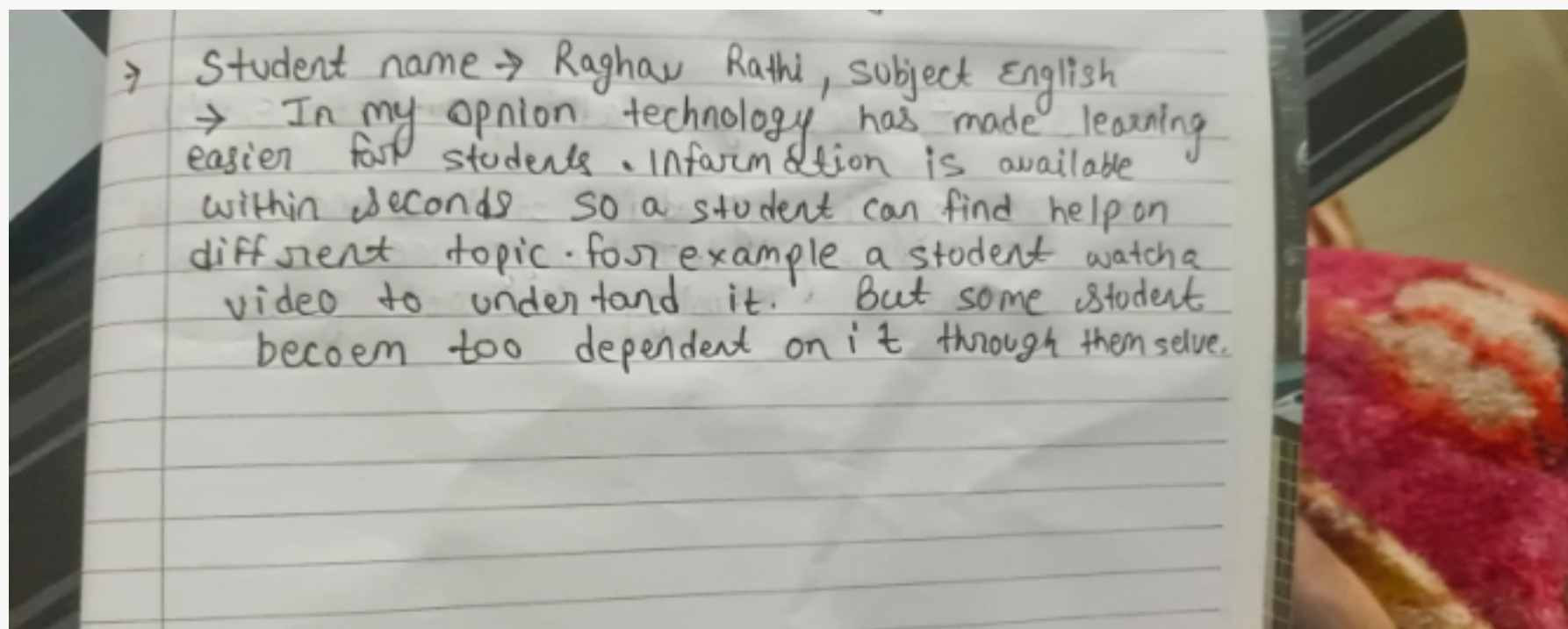
TEACHER NOTES

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+ Add Manual Note

5. OCR-like Spelling Errors — the original handwritten photo

Shown for direct reference — exactly what GradeSense actually read to produce the result on the previous page.



6. Unsupported File Upload

PASS

Why this test matters: Not one of the 8 required categories, but a real reliability gap worth covering: real photographed handwriting (PNG/JPG) is now a supported input, so this test proves a genuinely unrelated file type still gets rejected safely and clearly, not silently accepted or crashed on.

What we did: Attempted to upload a .gif file — not one of the supported formats — as the answer paper.

What happened: The upload was rejected immediately with a specific, visible error message listing the actual supported formats (PDF, DOCX, PNG, JPG) — no crash, no silent failure, no orphaned file left on the server.

GradeSense | Exam Grading Tool

Grade a Paper | History (1)

Grade a Student Answer Paper

Pick the question, provide the student's answer, and grade it — that's the whole flow.

1. EXAMINATION QUESTION

[English] Question 2 — Technology in Education Essay (5 Marks) ▾

+ Add Question

> View rubric criteria (5 Marks Available)

[Try a sample answer for this question](#)

2. STUDENT DETAILS (optional)

STUDENT NAME

Raghav Rath

ROLL NUMBER

01

3. ANSWER PAPER

Upload File

Paste Text

Drop a PDF, DOCX, or a photo of a handwritten answer

Unsupported file type — upload a PDF, DOCX, PNG, or JPG file, or use "Paste Text" instead.

Upload the answer paper (or switch to Paste Text) to continue

Grade This Paper →

7. Model / API Failure

PASS

Why this test matters: Assignment requires a test for "a model/API failure" — if the live grading API is unreachable, the system must not crash or hang; it must recover and say plainly that it did. This specifically tests the grading call, not transcription, so the same real answer (Raghav Rathi's handwritten essay, already transcribed in Case 1) was submitted as text rather than re-uploading the photo — otherwise a broken credential would fail transcription first and never reach the grading step being tested here.

What we did: Temporarily corrupted the live Gemini credentials, then submitted the same real answer for grading.

What happened: The system retried once, then automatically fell back to its deterministic offline grader, returned a real result (4/5), and marked the status as "degraded" with an explicit reason: "primary LLM API failed after a retry." Credentials were restored immediately afterward and live grading was re-verified to be working normally again.

Question 2 — Technology in Education Essay

Student: Raghav Rathi (Roll 01) · Subject: English

4 / 5

Needs Review

Graded in degraded mode — primary LLM API failed after a retry; results are from the fallback mock grader.

Mark as Reviewed

Student's Answer

Model Answer

GradeSense Answer Paper

Student: Raghav Rathi · Roll: 01

Subject: English
Date: 9/2/2026

Question 2 — Technology in Education Essay

Name- Raghav Rathi

In my opinion technology has made learning Signi Ficantly better for students. Information that used to take hour to find in a library is now avialable within seconds through search engine on line Coureses and educational vidoes. This means Students can explore topics at their own pace and revisit material as many times as they need until they truly understand it. for example as student who struggles with a difficult chemistry concept can watch several different explanation on Youtube until one of them finally makes sense.

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Even so I believe benefits outweigh this risk as long as technology is used thought fully. A student who uses the internet to understand a concept more deeply rather then to copy a finished answer, ends up learning more. Technology should be treated as a tool not ad replacement.

RUBRIC CRITERIA

SCORE: 4 / 5

> States a clear, unambiguous position on the essay prompt	1/1
> Provides well-structured arguments supported by relevant concrete examples	1/1
> Meaningfully considers and addresses an opposing viewpoint	1/1
> Uses relevant examples and demonstrates reasoning rather than merely making unsupported claims	1/1
> Concludes with a coherent, logical synthesis of the presented arguments	0/1

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